

- Q.27 Draw the key diagram of an 11kV substation.
- Q.28 Differentiate between Power wiring and Control wiring.
- Q.29 List the materials required for Plate Earthing.
- Q.30 Explain the Forwarding' operation of a 3 phase induction motor.
- Q.31 What is the role of a 'selected feeder' in motor supply?
- Q.32 Define Earth Electrode.
- Q.33 Explain the term 'Interlocking' in contactor circuits
- Q.34 Draw the schematic symbol for a 3 phase Induction Motor.
- Q.35 Why is charcoal used in an earthing pit?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw the Schematic and Power diagram for a DOL Starter for a 3 phase induction motor.
- Q.37 Explain Plate Earthing in detail with a neat labeled diagram.
- Q.38 Draw the layout of a Distribution Transformer earthing and list the materials needed.

(720) (4) 180944/170944/
120944/030944

No. of Printed Pages : 4 180944/170944/
Roll No. 120944/030944

4th Sem /

Elect., Power Station Engg., Elect. & Eltx. Engg.
Sub : Electrical Engineering Design & Drawing-II

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the full form of a DOL starter?
- a) Direct on line b) Dual on line
c) Direct off line d) Duel operated line
- Q.2 Which material is commonly used for earth electrodes?
- a) Plastic b) Copper/GI
c) Wood d) Rubber
- Q.3 A contactor is operated by:
- a) Mechanical lever b) Electromagnetic coil
c) Manual switch d) Rod
- Q.4 The purpose of earthing is to provide protection against:
- a) Over voltage b) Electric shock
c) High temperature d) High power
- Q.5 In a Star-Delta starter, the motor starts in :
- a) Delta b) Star
c) Series d) Parallel

(1) 180944/170944/
120944/030944

- Q.6 In a schematic diagram, 'NO' stands for:
- Not open
 - Normally open
 - New operation
 - Normally off
- Q.7 The standard frequency of AC supply in India is:
- 60Hz
 - 50Hz
 - 100Hz
 - 25Hz
- Q.8 Earth resistance can be reduced by adding:
- Sand
 - Charcoal and Salt
 - Stones
 - Concrete
- Q.9 CAD stands for:
- Computer Aided Design
 - Control and Design
 - Circuit Aided Drawing
 - Computer Aided Drawing
- Q.10 Which relay is used for sequential operation of two motors?
- Thermal Relay
 - Time Delay Relay
 - Overload Relay
 - Distance relay

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define the term 'Contactor'.
- Q.12 Mention one method of reducing earth resistance.

(2) 180944/170944/
120944/030944

- Q.13 Draw the symbol for a Normally Open (No) contact.
- Q.14 What is the primary used of a "Timer" in motor control?
- Q.15 Name the two types of earthing.
- Q.16 The wire used for earthing is called the _____ conductor.
- Q.17 True or False: Pipe earthing is better than plate earthing for dry soil.
- Q.18 What is the voltage level of a primary distribution transformer?
- Q.19 Name one graphic package used for drawing electrical circuits.
- Q.20 What does "IS" stand for in IS specifications?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the concept and purpose of earthing.
- Q.22 List any four components of a contactor control circuit.
- Q.23 Draw a neat labeled sketch of Pipe Earthing.
- Q.24 Describe the working of a Limit Switch.
- Q.25 What are the advantages of using CAD for electrical drawings?
- Q.26 Briefly explain the 'sequential operation' of two motors.

(3) 180944/170944/
120944/030944